Demo Trunk Line

STORM SEWER ANALYSIS REPORT

Project No. SS-DEMO-001 · Design Engineer, PE · Sample Engineering · City of Example · August 26, 2026

Design Basis

IDF i = 60.0/(t + 10.0)^0.80 in/hr   Design storm 10-yr   Units U.S. customary

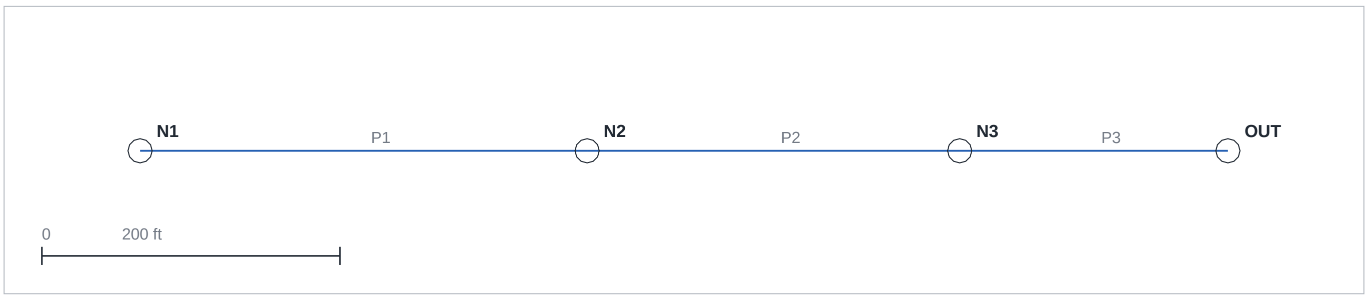
Network: 4 structures, 3 pipes, tailwater 100.50 ft

All pipes flow open-channel; no surface flooding.

Design Review

[WARN] P2 — Pipe P2: 89% of full capacity (> 85%)

Plan



Pipe Schedule

Pipe	From	To	Size in	Slope ft/ft	Tc min	i in/hr	Q cfs	Cap cfs	Full %	Vel ft/s	HGL dn ft	Status
P1	N1	N2	15	0.0050	12.0	5.06	3.54	4.58	77	4.12	103.79	ok
P2	N2	N3	18	0.0052	13.2	4.85	6.79	7.60	89	4.86	102.23	ok
P3	N3	OUT	21	0.0067	14.1	4.71	8.48	12.97	65	5.75	100.50	ok

Structure Schedule

Node	Tc min	Rim ft	HGL ft	Freeboard ft	Status
N1	12.0	110.00	104.96	5.04	ok
N2	13.2	108.50	103.79	4.71	ok
N3	14.1	107.00	102.23	4.77	ok
OUT	14.6	106.00	100.50	5.50	ok

Inlet Schedule (HEC-22, with bypass carryover)

Inlet	Local cfs	C/O in cfs	Intercepted cfs	Bypass cfs	To	Spread ft	Status
N1	3.82	0.00	1.42	2.40	(off)	11.9	EXCEEDS
N2	3.82	0.00	1.42	2.40	(off)	11.9	EXCEEDS

Pipe design flows remain full Rational C·A (conservative); this schedule checks surface capture and spread.

Profile (main trunk)